Video Transcript

Video 3: Using K-means to Classify Digits (05:09)

Now, we're going to apply K-Means to recognizing digits. So, there's a dataset of digits from zero through nine, and they're eight pixels by eight pixels. So, there's 64 pixels in each image, and each pixel has a grayscale number between and including 15. So, there's 16 Shades of grey. And we're going to load up this data and try and find out which images are similar to other images or which images are different. So, we need to have a metric of distance, D, between two images, A and B. So, say images eight and nine, we'd need to understand what's the difference. So, we're going to have to look at the grayscale color and take that into account. Now sklearns K-Means algorithm does a pretty good job of making a choice about that, but it's, as I say, it's not obvious how you trade off space with color. So, we load up the digits, the images, and we find there are 1797 of these images. Each image has 64 pixels. Each of those pixels has zero to 15 grayscale. So, each pixel, in some sense you can imagine, each pixel having a depth to it. Sometimes we talk about the the depth of the color, and here it's four bits, it's zero to 16.

And now, we'll apply the K-Means algorithm. We're going to choose 10 clusters because we know there are 10 digits, zero through nine. We ask K-means to cluster them, and the result is that we have 10 clusters. And so, let's take the center of each of those clusters. And so, that's a point in that pixel space if you like. And so, it'll look like an eight by eight image. So, here we plot out those centers. You can see most of the images there, one, two is a little, little shaky on two, and seven maybe a one, and eight here on the right is, that could be confused with the one. But you can see it does a surprisingly good job with this K-Means cluster. And in fact, here we're labeling each of these clusters. Now, it's arbitrary. Obviously, the cluster doesn't know what image it's looking at, it's just comparing grayscale and saying these are all close to each other.

Once we assign a label to a cluster, you can see that given any other image, we have about an 80% chance of identifying it, 0.795, so that's the accuracy score of given an image. We can label it and we'll be right 80% of the time. And we can also look at the confusion matrix. Now, we may not have come across this yet, so we've got 10 clusters, so zero through nine, and we're plotting which ones get confused. And you can see here, eight and one get confused. It has a score of 99.

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Five and nine get confused 42 times, eight and nine, 48. So, you see that there is some confusion and you see other ones here down at seven, quite low. But most of the time there is no confusion that we're identifying the image correctly. So, that's applying K-Means and you can see that it's quite a powerful algorithm and did a very good job with identifying these images. So, that's how you can apply it. So, K-Means clustering.

Why don't you run through these algorithms and make sure you understand them? There's some very neat Python structures in there, so take a look at those. In particular, I'm asking you to take a look at how this works. So, I want you to look up what this does, this permutation on random state, understand and take a look at what that i is. What's it doing? So, this is in find_clusters. I'll give you all this code and I hope you understand K-Means, it's quite a simple algorithm, but quite powerful. Okay, bye for now.

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